Statistical Learning with Extreme Values

Presentation - Paper Reading Assignment

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Sparse representation of multivariate extremes with applications to anomaly detection

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MATHÉMATIQUES VISION APPRENTISSAGE

The "Crypto" Intuition: When is a Crash an Anomaly?

The Scenario: You manage a portfolio of 6 assets: `BTC`, `ETH`, `BNB`, `SOL`, `XRP`, and `USDT` (Stablecoin).

Two types of Extreme Events:

Type A: Systemic Crash

Bitcoin drops 20%, everything else drops 20%.

Is this an anomaly?
NO. It's a standard market crash.

Type B: The De-peg

Bitcoin is stable, but **USDT** (Stablecoin) drops 20%.

Is this an anomaly?
YES. This is structural failure.

The Challenge: Standard anomaly detection (Distance/Correlation) fails here because both events look "far from the mean." We need **Multivariate EVT**.

The Theoretical Barrier: The Curse of Dimensionality

Goal: Model the dependence of extremes (The Angular Measure Φ).

The Problem

- In high dimensions ($d > 10$), the "sphere" of directions is too big.
- Data is sparse: We cannot estimate the density everywhere.

The Solution (Goix et al., 2017) [GSC17]: Sparsity Hypothesis

- Extremes don't happen "everywhere" on the sphere.
- They concentrate on specific low-dimensional **sub-cones** (faces).

Intuition: In our data, a crash of $\{BTC, ETH\}$ is common.
A crash of $\{BTC, USDT\}$ is physically unlikely.

Geometric Approach: ϵ-Thickened Rectangles

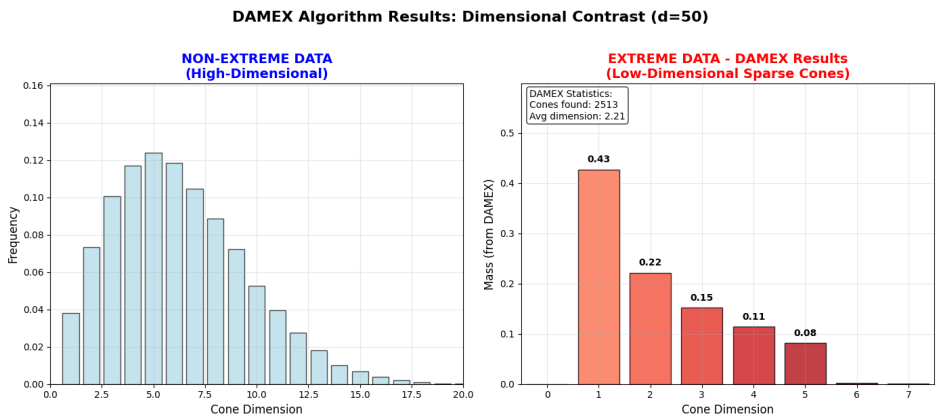
Problem: Real data never lies *exactly* on a lower-dimensional face.

Defining a "Cone" (Pattern)

An observation x belongs to pattern α if:

Active ($j \in \alpha$): Large ($> \epsilon \|x\|_\infty$)

Inactive ($j \notin \alpha$): Small ($\leq \epsilon \|x\|_\infty$)



The Estimator: We count how many extreme points match this pattern.

$$\hat{\mathcal{M}}(\alpha) = \frac{1}{k} \sum_{i=1}^{n} \mathbb{I}\left\{\text{Point } i \in \text{Cone } \alpha \cap \{R_i > n/k\}\right\}$$

Note: We apply a threshold to remove cones with negligible mass (noise).

DAMEX: Detecting Anomaly with Multivariate EXtremes

1. **Standardize:** Transform data to Pareto scale

$$S_j(x) = \#\{X_i^j > x, i \leq n\}/n \quad V_i^j = 1/S_j(X_i^j)$$

2. **Learn:** Estimate the mass $\hat{\mathcal{M}}(\alpha)$ for every cone (The "Normal" Profile).
3. **Score:** For a new extreme point x:

The Anomaly Score

$$s_n(x) \propto \frac{\text{Mass of the Cone containing } x}{\text{Magnitude of } x}$$

- **Systemic Crash:** $x \in \{BTC, ETH, ...\}$. This cone has **High Mass**. $\Rightarrow$ Score is High (Normal Extreme).
- **De-peg:** $x \in \{BTC, USDT\}$. This cone has **Zero Mass**. $\Rightarrow$ Score is Low (Anomaly).

Feature Clustering Methodology

Goal: Find features with similar behavior on extreme data.

Dependence test (cluster $A = \{i, j, k\}$, extreme data only):

$$H_0 : P(\mathrm{Ex}) = \prod P(\mathrm{Ex}_i)$$

$$H_1 : P(\mathrm{Ex}) > \prod P(\mathrm{Ex}_i)$$

Optimization: Use DAMEX cones, minimize:

$$\mathrm{AIC} = \text{Training Deviance} + 2 \times \frac{\#\text{subfaces}}{n_{\text{extremes}}}$$

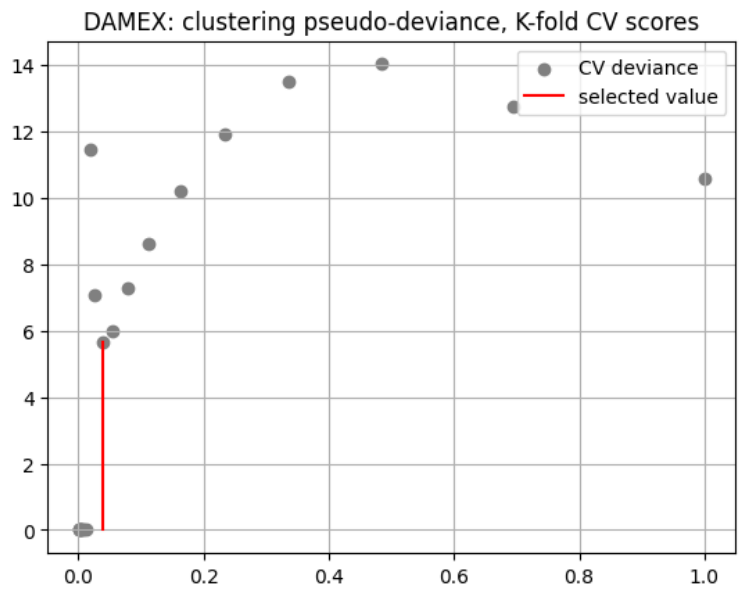


Figure: AIC vs ϵ: 1) Null model, 2) Optimal, 3) Overfit, 4) Double descent

The Structure of Crypto Crashes

We applied DAMEX to daily returns (2020-2024). **What "Normal" extremes did it find?**

Cone Type	Assets Involved	Mass (Likelihood)
1. Systemic	BTC + ETH + BNB + SOL + XRP *(The whole market moves together)*	**1.00**
2. Idiosyncratic	USDT (Stablecoin) *(Stablecoin volatility is isolated)*	**0.72**
3. Noise	SOL + USDT *(Rare/Weak dependence)*	**0.18**

Finding: The algorithm successfully "decoupled" the Stablecoin from the speculative assets without supervision.

Conclusion

Summary:

1. **Sparsity is key:** In high dimensions, extremes concentrate on low-dimensional faces (sub-groups).
2. **Geometry over Magnitude:** Anomalies in the tail are defined by *directions* (ϵ-cones), not just distance.
3. **Practical Value:** DAMEX allows risk managers to distinguish "Panic Selling" (Systemic) from "Asset Failure" (Structural).

Future Work:

- Selecting ϵ and k is hard. We implemented an AIC criterion (see report) to automate this.

Bibliography I

[GSC17] Nicolas Goix, Anne Sabourin, and Stephan Clémençon. "Sparse representation of multivariate extremes with applications to anomaly detection". In: *Journal of Multivariate Analysis* 161 (2017), pp. 12–31. ISSN: 0047-259X. DOI: https://doi.org/10.1016/j.jmva.2017.06.010. URL: https://www.sciencedirect.com/science/article/pii/S0047259X17304062.

Appendix